Geometric Purity and Multi-Scale Diagnostic Ratios

Luís Pimenta Damásio

Independent Researcher, Lisbon, Portugal • ORCID: 0009-0001-7606-7557

ABSTRACT

To analyze physical systems spanning over one hundred orders of magnitude without losing structural clarity in raw dimensional units, we construct four dimensionless diagnostic invariants: **Weyl Conformal Purity** (χ), the **Gravity-to-Thermal Decoherence Crossover Ratio** ($\mathscr{R}_{\mathrm{DP}}$), **Penrose–Zel'dovich Rotational Extraction Gain** ($\mathscr{E}_{\mathrm{Penrose}}$), and the **Semiclassical Breakdown Index** (Ξ_{QG}). Derived from the Seeley–DeWitt a_4 trace anomaly, Buckingham-Π dimensional analysis, Novikov–Thorne binding energies, and the L_∞ Chebyshev supremum norm, these metrics separate geometric vacuum polarization from environmental noise while establishing strict epistemic boundaries for semiclassical gravity.

Keywords: Weyl trace anomaly, Penrose–Diósi collapse, Buckingham-Π theorem, Novikov–Thorne efficiency, Semiclassical breakdown, Dimensionless invariants

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1. First-Principles Mathematical Derivations

1.1 Weyl Conformal Purity (χ) via Seeley–DeWitt Trace Anomaly

In classical massless field theory, the stress-energy tensor is traceless ($T^{\mu}{}_{\mu} = 0$) due to invariance under local Weyl rescalings ($g_{\mu\nu} \to \Omega^2(x)\, g_{\mu\nu}$). In curved spacetime, quantum vacuum fluctuations break this symmetry, generating the **Weyl Conformal Trace Anomaly** ($\langle T^{\mu}{}_{\mu} \rangle_{\mathrm{ren}}$) governed by the one-loop Seeley–DeWitt a_4 heat-kernel coefficient:

$$\mathcal{A}_{\mathrm{W}} \equiv \langle T^{\mu}{}_{\mu} \rangle_{\mathrm{ren}} / \rho_{\mathrm{vac,ref}} = \mathcal{C}_{\mathrm{clamped}}{}^2 / (60\pi^2)$$

where $\mathcal{C}_{\mathrm{clamped}} = \min(\mathcal{C}_{\mathrm{hor}}, \mathcal{C}_{\mathrm{eff}})$ bounds the dimensionless compactness at the Kerr–Newman horizon $\mathcal{C}_{\mathrm{hor}} = 2 / (1 + \sqrt{(1 - a_*{}^2 - Q_*{}^2)})$. Normalizing $\mathcal{A}_{\mathrm{W}} \geq 0$ by the characteristic one-loop phase-space factor (60) yields the bounded **Weyl Conformal Purity**:

$$\chi \equiv 1 / (1 + 60\mathcal{A}_{\mathrm{W}}) = 1 / (1 + \mathcal{C}_{\mathrm{clamped}}{}^2 / \pi^2) \in (0, 1]$$

- **Minkowski Limit ($\mathcal{C}_{\mathrm{clamped}} = 0$):** $\mathcal{A}_{\mathrm{W}} = 0 \Rightarrow \chi = 1.0000$.
- **Schwarzschild Horizon ($\mathcal{C}_{\mathrm{clamped}} = 1$):** $\chi = \pi^2 / (\pi^2 + 1) \approx 0.9080$.
- **Extremal Kerr–Newman Horizon ($\mathcal{C}_{\mathrm{clamped}} \to 2$):** $\chi = \pi^2 / (\pi^2 + 4) \approx 0.7116$.

1.2 Gravity-to-Thermal Crossover Ratio ($\mathscr{R}_{\mathrm{DP}}$) via Buckingham-Π Scaling

Comparing the Penrose–Diósi gravitational self-energy collapse rate $\Gamma_{\mathrm{grav}} = (M_{\mathrm{eff}} / M_{\mathrm{P}})^2 \, (c / L)$ against the Caldeira–Leggett thermal photon scattering rate $\Gamma_{\mathrm{th}} = L^2 \, (k_{\mathrm{B}} \, T_{\mathrm{eff}})^3 / (\hbar^3 \, c^2)$ eliminates time units completely, forming a dimensionless Buckingham-Π group:

$$\mathscr{R}_{\mathrm{DP}} \equiv \Gamma_{\mathrm{grav}} / \Gamma_{\mathrm{th}} = (M_{\mathrm{eff}} / M_{\mathrm{P}})^2 \cdot (\hbar^2 \, c^3) / [L^3 \, (k_{\mathrm{B}} \, T_{\mathrm{eff}})^3] = (M_{\mathrm{eff}} / M_{\mathrm{P}})^2 \cdot (\lambda\!\!\!^{-}_{\mathrm{th}} / L)^3$$

where $\lambda\!\!\!^{-}_{\mathrm{th}} = \hbar c / (k_{\mathrm{B}} \, T_{\mathrm{eff}})$ is the reduced thermal wavelength. The critical crossover $\mathscr{R}_{\mathrm{DP}} = 1$ occurs along the mesoscopic mass-temperature boundary $M_{\mathrm{eff}} / M_{\mathrm{P}} = (L / \lambda\!\!\!^{-}_{\mathrm{th}})^{3/2}$.

1.3 Penrose–Zel'dovich Rotational Extraction Gain ($\mathscr{E}_{\mathbf{Penrose}}$)

To quantify rotational energy extraction from a spinning Kerr–Newman ergosphere at polar angle θ, we combine three dimensionless physical factors:

1. **Zel'dovich–Starobinsky Superradiant Wave Gain:** $\mathcal{G}_{\mathrm{SR}} = \max(0, 1 - f_{\mathrm{local}} / f_{\mathrm{H}})$, where $f_{\mathrm{H}} = c \, a_* / (4\pi \, r_+)$ is the horizon angular frequency.
2. **Polar Ergosphere Coupling:** $\sin^2\theta$, reflecting the vanishing of ergosphere thickness along the rotation axis (θ = 0°) and peak frame-dragging at the equator (θ = 90°).
3. **Novikov–Thorne Radiative Binding Efficiency:** $\eta_{\mathrm{NT}}(r_{\mathrm{ISCO}}) = 1 - \sqrt{(1 - 2r_{\mathrm{g}} / (3 \, r_{\mathrm{ISCO}}))}$, ranging from 5.72% (Schwarzschild) to 42.26% (extremal Kerr).

Taking their product defines the **Penrose–Zel'dovich Extraction Gain**:

$$\mathscr{E}_{\mathrm{Penrose}}(\theta) \equiv \max(0, 1 - f_{\mathrm{local}} / f_{\mathrm{H}}) \cdot \sin^2\theta \cdot \eta_{\mathrm{NT}}(r_{\mathrm{ISCO}})$$

1.4 Semiclassical Breakdown Index ($\Xi_{\mathbf{QG}}$) via L_∞ Supremum Norm

Semiclassical Einstein gravity ($G_{\mu\nu} = (8\pi G / c^4) \, \langle \hat{T}_{\mu\nu} \rangle_{\mathrm{ren}}$) holds only while four independent dimensionless ratios remain strictly below unity:

- **Minimal Planck Length Cutoff:** $\Pi_1 = \ell_{\mathrm{P}} / L$
- **Stretched-Horizon Membrane Cutoff:** $\Pi_2 = 10^{-6} / N^2(\theta)$
- **Sub-Extremal Cosmic Censorship Cutoff:** $\Pi_3 = (a_*^2 + Q_*^2) / 0.9998$
- **Planck Thermal Foam Ceiling:** $\Pi_4 = T_{\mathrm{eff}} / T_{\mathrm{P}}$

Taking the L_∞ (Chebyshev) supremum norm defines the **Semiclassical Breakdown Index**:

$$\Xi_{\mathrm{QG}} \equiv \|(\Pi_1, \Pi_2, \Pi_3, \Pi_4)\|_\infty = \max(\ell_{\mathrm{P}} / L, \; 10^{-6} / N^2(\theta), \; (a_*^2 + Q_*^2) / 0.9998, \; T_{\mathrm{eff}} / T_{\mathrm{P}})$$

2. Plain-Language Physical & Operational Guide

- **Weyl Conformal Purity (χ):** Measures how severely spacetime curvature warps the quantum vacuum even at absolute zero and zero elapsed time. It equals 1.0000 in flat space and drops toward 0.908 at a Schwarzschild horizon or 0.712 at an extremal Kerr–Newman horizon.
- **Gravity-to-Thermal Crossover ($\mathscr{R}_{\mathbf{DP}}$):** Acts as a dimensionless tug-of-war gauge between environmental thermal decoherence ($\mathscr{R}_{\mathrm{DP}} < 1$) and Penrose–Diósi gravitational self-energy collapse ($\mathscr{R}_{\mathrm{DP}} > 1$), with $\mathscr{R}_{\mathrm{DP}} = 1$ marking the exact quantum-to-classical mesoscopic threshold.
- **Penrose–Zel'dovich Extraction Gain ($\mathscr{E}_{\mathbf{Penrose}}$):** Expresses the net fractional energy extractable from a rotating ergosphere at latitude θ, vanishing identically along the spin axis ($\theta = 0°$) and above the superradiant frequency cutoff ($f_{\mathrm{local}} \geq f_{\mathrm{H}}$).
- **Semiclassical Breakdown Index ($\Xi_{\mathbf{QG}}$):** Serves as a continuous epistemic boundary monitor. Because it uses the supremum (max) norm, $\Xi_{\mathrm{QG}} \to 1.0000$ the instant *any single* semiclassical assumption reaches its physical breakdown wall.

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